

Examination #3

NAME: Solution

SHOW ALL WORK

- 1) Link AB of the linkage mechanism shown rotates with an instantaneous counterclockwise angular velocity of 10 radians/second. What is the instantaneous angular velocity of link BC when link AB is horizontal and link CD is vertical? (SHOW WORK AND USE METHOD DISCUSSED IN CLASS)

- a) 2.25 radians/second (CW)
b) 3.25 radians/second (CCW)
c) 5.50 radians/second (CW)
d) 12.5 radians/second (CW)

If ω_{AB} is $\uparrow$
 ω_{BC} must be $\downarrow$
Get rid of (B)

$$V_B = V_A + \omega_{AB} \times r_{A/B}$$

$$V_B = (r_{A/B})(\omega_{AB})$$

$$V_B = (5)(10) = 50 \text{ m/s} \uparrow$$

$$V_C = V_B + \omega_{BC} \times r_{B/C}$$

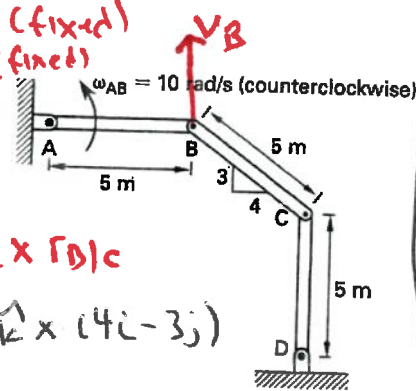
$$-V_C \hat{i} = 50 \hat{j} + \omega_{BC} \hat{k} \times (4\hat{i} - 3\hat{j})$$

j analysis

$$0 = 50 + 4\omega_{BC}$$

$$4\omega_{BC} = -50$$

$$\omega_{BC} = 12.5 \text{ rad/s} \downarrow (\omega = -)$$



$\hat{i} \times \hat{j} = \hat{k}$
 $\hat{j} \times \hat{k} = \hat{i}$
 $\hat{k} \times \hat{i} = \hat{j}$
 $\hat{i} \times \hat{k} = -\hat{j}$
 $\hat{k} \times \hat{j} = -\hat{i}$
 $\hat{j} \times \hat{i} = -\hat{k}$

- 2) Two 2 kg blocks are linked as shown. Assume the surfaces are frictionless. What is the velocity of block B if block A is moving at a speed of 3 m/s? (SHOW WORK AND USE METHOD DISCUSSED IN CLASS)

- a) 0 m/s
b) 1.30 m/s
c) 1.73 m/s
d) 5.20 m/s

$$V_B = V_B \hat{j}$$

$$V_A = 30 \frac{\text{m}}{\text{s}} \hat{i} = \text{Given}$$

$$V_B = V_A + \omega_{BC} \times r_{A/B}$$

$$V_B \hat{j} = 3 \hat{i} + \omega_{BC} \hat{k} \times (3 \cos 30^\circ \hat{i} + 3 \sin 30^\circ \hat{j})$$

$$V_B \hat{j} = 3 \hat{i} + 2.59 \omega_{BC} \hat{j} - 1.5 \hat{i} \omega_{BC}$$

i analysis

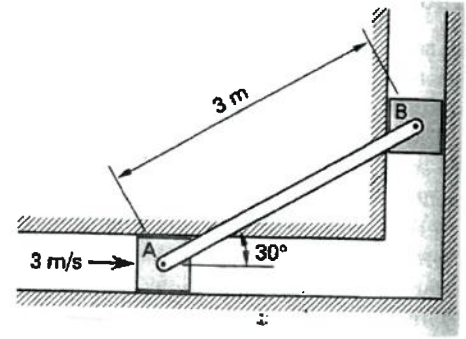
$$0 = 3 - 1.5 \omega_{BC}$$

$$\omega_{BC} = 2 \text{ rad/s}$$

j analysis

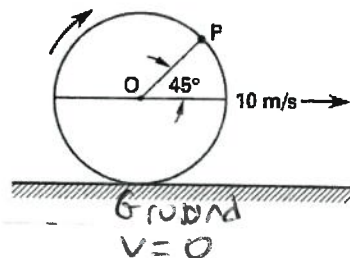
$$V_B = (2.59)(2) = 5.18 \text{ m/s}$$

$$V_B = 5.2 \text{ m/s}$$



3. A disk rolls along a flat surface at a constant speed of 10 m/s. Its diameter is 0.5 m. At a particular instant, point P on the edge of the disk is 45° from the horizontal. What is the velocity of point P at that instant? SHOW WORK AND USE METHOD DISCUSSED IN CLASS.

- a) 10 m/s
- b) 15 m/s
- c) 16.2 m/s
- d) 18.5 m/s



Given
 $d = 0.5 \text{ m}$
 $r = 0.25 \text{ m}$
 $v_{\text{center}} = 10 \text{ m/s}$
 $v = r\omega$ and so
 $\omega = \frac{v}{r} = \frac{10}{0.25} = 40 \frac{\text{rad}}{\text{s}}$

$$v_p = 0 + 40 \hat{k} \times (0.25 \cos 45^\circ \hat{i} + 0.25 \sin 45^\circ \hat{j} + 0.25 \hat{j})$$

$\downarrow$
Ground

$$v_p = (7.07 \hat{j} + 17.07 \hat{i})$$

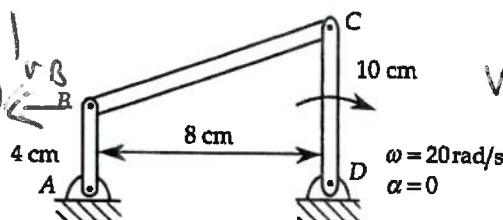
$$v_p = 18.5 \text{ m/s}$$

4. Find ω_{AB} in radians/second. SHOW ALL WORK AND USE METHOD DISCUSSED IN CLASS.

- a) 20 ccw
- b) 20 cw
- c) 10 ccw
- d) 10 cw

e) 0

$v_A = 0$ (fixed)
 $v_D = 0$ (fixed)



$v_C = (20)(10)$
 $= 200 \frac{\text{cm}}{\text{s}}$

(1) $v_B = v_C + \omega_{BC} \times (r_{B/C})$
 $v_B = -200 \hat{i} + \omega_{BC} \hat{k} \times (8 \hat{i} + 6 \hat{j})$
 $\omega_{BC} = 33.3 \text{ rad/s}$

(2) $v_A = v_B + \omega_{AB} \times (r_{A/B})$
 $v_A = 266 \frac{\text{cm}}{\text{s}} \hat{i} + \omega_{AB} \hat{k} (4 \hat{i} + 0 \hat{j})$
 $v_A = 266 \hat{i} + 4 \omega_{AB} \hat{j}$

STUCK $\rightarrow$ CANNOT SOLVE $\omega_{AB} = 0$

Also

$v_{A/B} = r_{AB} \omega_{AB} \perp AB$

v_B is horizontal

$v_{A/B} = v_A - v_B$

$v_{A/B} = 0$
 and
 so
 $\omega_{AB} = 0$

5) What is the force at O immediately after the string is cut? SHOW ALL WORK.

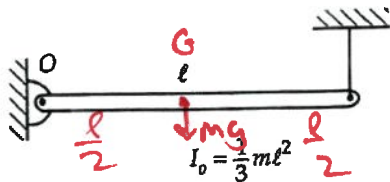
a) $mg/2$

b) $mg/3$

c) $mg/4$

d) $mg/5$

e) 0



$$\sum M_O = I_O \alpha$$

$$mg \times \frac{L}{2} = \frac{1}{3} mL^2 \alpha$$

$$\alpha = \frac{3g}{2L} \quad (1)$$

$$\sum F_y = ma_y = m r \alpha$$

$$mg - F_{\text{force at O}} = m \left(\frac{L}{2} \right) \alpha \quad (2)$$

Substitute (1) into (2)

$$F_O = mg - m \left(\frac{L}{2} \right) \left(\frac{3g}{2L} \right)$$

$$F_O = mg - \frac{3mg}{4} = \frac{mg}{4}$$

Questions 6 and 7 refer to the following.

$$S(t) = 2 \sin t \hat{i} + 4 \cos t \hat{j} \quad (t \text{ in radians})$$

6) What is the magnitude of the particle's velocity at $t = 4$ radians? SHOW ALL WORK

a) 2.61

b) 2.75

c) 3.30

d) 4.12

$$V(t) = \frac{ds}{dt} = 2 \cos t \hat{i} - 4 \sin t \hat{j}$$

$$V(4) = 2 \cos(4 \text{ radians}) \hat{i} - 4 \sin(4 \text{ radians}) \hat{j}$$

$$V(4) = -1.31 \hat{i} - (-3.03) \hat{j}$$

$$V = 3.30 \text{ m/s}$$

7) What is the magnitude of the particle's acceleration at $t = \pi$? SHOW ALL WORK

a) 2

b) 2.56

c) 3.14

d) 4.00

$$a(t) = \frac{dv}{dt} = -2 \sin t \hat{i} - 4 \cos t \hat{j}$$

$$a(\pi) = -2 \sin \pi \hat{i} - 4 \cos \pi \hat{j}$$

$$a(\pi) = 0\hat{i} + 4\hat{j}$$

$$a(\pi) = \sqrt{(0)^2 + (4)^2}$$

$$a = 4 \text{ m/s}^2$$

8) The rotor of a steam turbine is rotating at 7200 revolutions/minute when the steam supply is suddenly cut off. The rotor decelerates at a constant rate and comes to rest after 5 minutes. What was the angular deceleration of the rotor? SHOW ALL WORK

a) 0.40 rad/s²

b) 2.5 rad/s²

c) 5.8 rad/s²

d) 16 rad/s²

$$t = 5 \text{ minutes} \times \frac{60 \text{ s}}{1 \text{ min}} = 300 \text{ s}$$

$$\omega = 7200 \frac{\text{revolutions}}{\text{min}} \times \frac{2\pi \text{ radians}}{1 \text{ revolution}} \times \frac{1 \text{ min}}{60 \text{ s}}$$

$$\omega_i = 754 \text{ rad/s} \quad \omega_f = 0 \text{ (STOP)}$$

$$\omega_{\text{final}} = \omega_i + \alpha t$$

$$\alpha = \frac{\omega_f - \omega_i}{t} = \frac{0 - 754}{300} = -2.5 \text{ rad/s}^2 \quad (\text{deceleration})$$

9) A flywheel rotates at 7200 rev/min when the power is suddenly cut off. The flywheel decelerates at a constant rate of 2.1 rad/s² and comes to rest 6 minutes later. How many revolutions does the flywheel make before coming to rest?

a) 18,000 rev

b) 22,000 rev

c) 72,000 rev

d) 390,000 rev

$$\theta = \theta_0 + \omega_0 t - \frac{1}{2} \alpha t^2$$

$$\theta =$$

$$7200 \frac{\text{rev}}{\text{min}} = 754 \text{ rad/s}$$

$$5 \text{ min} \times \frac{60 \text{ s}}{1 \text{ min}} = 300 \text{ s}$$

$$\theta = 754(360) - \frac{1}{2}(2.1)(360)^2$$

$$271440 - 136080$$

$$\theta = 135360 \text{ radians} \times \frac{1 \text{ rev}}{2\pi \text{ rad}} = 21554 \text{ rev}$$

10) A uniform rod (AB) of length L and weight W is pinned at point C and restrained by cable OA. The cable is suddenly cut. The rod starts to rotate about point C with point A moving down and point B moving up. What is the linear acceleration of point B? SHOW ALL WORK

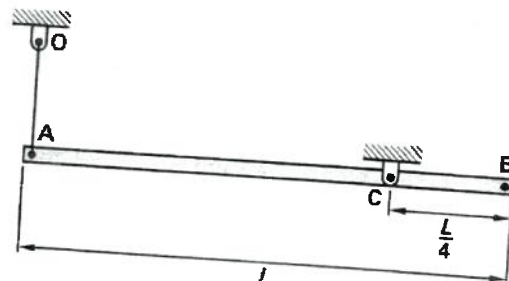
a) $3g/16$

b) $g/4$

c) $3g/7$

d) $3g/4$

$I_C = I_{C0} + m d^2$
(Parallel Axis Theorem)



$$I_C = \frac{1}{12} mL^2 + m\left(\frac{L}{4}\right)^2$$

$$I_C = mL^2\left(\frac{1}{12} + \frac{1}{16}\right)$$

$$I_C = \left(\frac{7}{48}\right) mL^2$$

$$\sum M_C = \left(\frac{3W}{4}\right)\left(\frac{3L}{4}\right) - \left(\frac{W}{4}\right)\left(\frac{L}{4}\right)$$

$$\sum M_C = \frac{WL}{4} = \frac{mgL}{4}$$

$$\sum M_C = I \alpha$$

$$\alpha = \frac{\frac{mgL}{4}}{\frac{7}{48} mL^2} = \frac{12g}{7L}$$

and so
 $a = r\alpha$

$$a = \left(\frac{L}{4}\right)\left(\frac{12g}{7L}\right)$$

$$a = \frac{3g}{7}$$

11)

Two blocks are joined by a string that passes over the pulley which has radius R and moment of inertia I about its center. T_1 and T_2 are the tensions in the string on either side of the pulley and α is the angular acceleration of the pulley. Which of the following equations best describe the pulley's rotational motion during the time the blocks accelerate? (SHOW HOW YOU CAME UP WITH YOUR FINAL ANSWER)

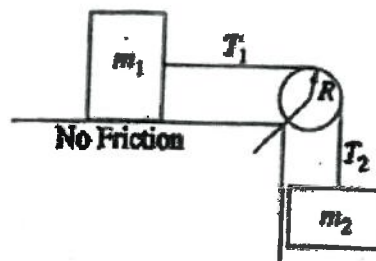
$$m_2 g R = I \alpha$$

$$(T_1 + T_2) R = I \alpha$$

$$T_2 R = I \alpha$$

$$(T_2 - T_1) R = I \alpha$$

$$(m_2 - m_1) g R = I \alpha$$



$\text{CW} = (-)$
 $\text{CCW} = (+)$

$$\sum M_C = I \alpha$$

$$T_2 R - T_1 R = I \alpha$$

$$(T_2 - T_1) R = I \alpha$$

13)

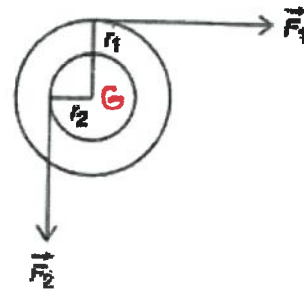
A solid cylinder consisting of an outer radius r_1 and inner radius r_2 is pivoted on a frictionless axle as shown. A string is wound around the outer radius and is pulled to the right with a force $F_1 = 3 \text{ N}$. A second string is wound around the inner radius and is pulled with a force $F_2 = 5 \text{ N}$. If $r_1 = 0.75 \text{ m}$ and $r_2 = 0.35 \text{ m}$, what is the net torque acting on the cylinder? (SHOW HOW YOU CAME UP WITH YOUR FINAL ANSWER)

a) 2.25 N m

b) -2.25 N m

c) 0.5 N m

d) -0.5 N m



$\text{CW} = (-)$

$\text{CCW} = (+)$

$$\sum M_G = -F_1 r_1 + F_2 r_2$$

$$\sum M_G = (-3)(0.75) + (5)(0.35)$$

$$\sum M_G = -2.25 \text{ N} \cdot \text{m} + 1.75 \text{ N} \cdot \text{m}$$

$$\sum M_G = \underline{\underline{-0.5 \text{ N} \cdot \text{m}}}$$

14) Two wheels are fixed to each other and are free to rotate about a frictionless axis through their concentric center. As shown below, 4 forces are exerted tangent to the wheels. The magnitude of the net torque is: (SHOW HOW YOU CAME UP WITH YOUR FINAL ANSWER)

A) ZERO

c) $2FR$

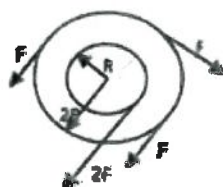
e) $8FR$

B) FR

d) $4FR$

$CW = [-]$

$CCW = [+]$



$$\begin{aligned}\sum \tau_o &= (F)(2R) - F(2R) - (2F)(R) - (F)(2R) \\ &= 2FR - 2FR - 2FR - 2FR \\ &= -4FR\end{aligned}$$

15. A rod of length L is rotated about its center which has a moment of inertia $(1/12) ML^2$. What is the moment of inertia at a point " $L/4$ " away from the center?

a) $(1/12) ML^2$

b) $(3/4) ML^2$

c) $(7/48) ML^2$

d) $(15/29) ML^2$

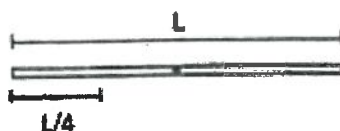
e) ML^2

$$I_{\text{final}} = I_o + mr^2 = \text{Parallel Axis Theorem}$$

$$r = \frac{L}{4}$$

$$I = \frac{1}{12} mL^2 + m\left(\frac{L}{4}\right)^2$$

$$I = \frac{1}{12} mL^2 + \frac{mL^2}{16}$$



$$I = \frac{28mL^2}{192}$$

$$I = \frac{7mL^2}{48}$$

$$\boxed{I = \frac{7}{48} mL^2}$$